

	MAMA NGINA UNIVERSITY COLLEGE QUALITY MANAGEMENT SYSTEM	Ref.:	MNUC/SPAS/CO/1.0
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Title	COURSE OUTLINE	Date:	2nd Sept 2026

UNIT CODE: SCO 203

UNIT TITLE: Software Testing and Quality Assurance

Lecturer: Dr David Kaimenyi – kaimenyi.david@mnu.ac.ke

Semester 1 (Sept – Dec 2026)

Prerequisite: SCO101

Course Description

Introduction to software testing. Unit Testing. Control Flow Testing. Data Flow Testing. System Integration Testing. System Test Categories. Functional Testing. System Test Design. System Test Planning and Automation. System Test Execution. Acceptance Testing. Software Inspection. Software Quality Assurance. Software Quality Standards.

Objective

This course examines the theory and practice behind software testing and quality assurance. Emphasis is placed on understanding the software testing process, planning, strategy, criteria, and testing methods, as well as software quality assurance concepts & control process.

Expected Learning Outcomes

By the end of this course, the student should be able to:

1. Discuss practices that support the production of quality software
2. Describe software testing techniques
3. Describe life-cycle models for requirements, defects, test cases, and test results
4. Explain process models for units, integration, system, and acceptance testing

Detailed Course Schedule

Week	Topic	Subtopics
1	Introduction to Software Testing	Definition and objectives of software testing; Importance of software quality; Principles of software testing; Testing vs. debugging; Software Testing Life Cycle (STLC)
2	Unit Testing	Concepts and objectives of unit testing; Unit test planning and design; Test drivers and stubs; Unit testing frameworks; Best practices in unit testing
3	Control Flow Testing	Introduction to white-box testing; Control Flow Graphs (CFGs); Statement coverage; Branch coverage; Path coverage; Cyclomatic complexity

Week	Topic	Subtopics
4	Data Flow Testing	Concepts of data flow testing; Definition-Use (DU) pairs; Data flow anomalies; Data flow graphs; Test case generation using data flow analysis
5	System Integration Testing	Integration testing concepts; Top-down integration; Bottom-up integration; Big Bang integration; Sandwich integration; Interface testing
6	CAT I	Continuous Assessment Test covering Weeks 1–5
7	System Test Categories and Functional Testing	System testing concepts; Functional testing; Non-functional testing; Regression testing; Performance testing; Security testing; Usability testing
8	System Test Design	Test design techniques; Equivalence partitioning; Boundary value analysis; Decision table testing; State transition testing
9	System Test Planning and Automation	Test planning process; Test strategy development; Test environment setup; Introduction to test automation; Automation tools and frameworks
10	CAT II	Continuous Assessment Test covering Weeks 7–9
11	System Test Execution and Acceptance Testing	Test execution process; Defect reporting and tracking; User Acceptance Testing (UAT); Alpha testing; Beta testing; Test closure activities
12	Software Inspection, Quality Assurance and Standards	Software inspections and reviews; Software Quality Assurance (SQA); Verification and Validation; Software quality models; ISO 9001; ISO/IEC 25010; IEEE software quality standards
13		Examination

Delivery of the Unit

Lectures: 2 Hours per week; Tutorials 2 hours per week.

Assessments

Regular Examination at end of semester; 70%, Continuous Assessment: 30%, where 20% shall be continuous assessment tests, 10% shall be assignments.

Textbooks and References

1. Naik, K., & Tripathy, P. (2011). Software testing and quality assurance: theory and practice. John Wiley & Sons.
2. Ainapure, B. S. (2009). Software testing and quality assurance. Technical Publications.
3. Agarwal, B. B., Tayal, S. P., & Gupta, M. (2010). Software engineering and testing. Jones & Bartlett Learning.
4. Fenton, N., & Bieman, J. (2014). Software metrics: a rigorous and practical approach. CRC Press.